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Paper Title: Gesture-Based Interactive Gaming Using MLP: A
Deep Learning Approach to Flappy Bird Control

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Motivation

- Hand Gesture Recognition (HGR) is fundamental to Human-Computer Interaction (HCI) — enables natural control for gaming, robotics, and VR [3]
- Traditional methods (keyboard/mouse) limit accessibility and immersion [10]
- Goal: A gesture-controlled Flappy Bird game using three gestures
 - **Thumbs Up** → Flap Up
 - **Thumbs Down** → Dive Down
 - **Neutral** → No action
- Proof-of-concept extendable to hospitals, assistive tech, and VR

Identified Gaps in Existing Work

➤ **Computational Inefficiency**

CNN/Transformer models require GPUs — unsuitable for lightweight real-time

➤ **Application Gaps**

Many existing systems remain research prototypes without being deployed for practical real-world applications. [10]

➤ **Lack of Interpretability**

No existing gesture-recognition systems use Explainable AI (XAI) techniques to verify whether the model learns meaningful hand-gesture patterns instead of relying on irrelevant or random features.

Problem Statement

Problem:

- Rule-based if/else methods are fragile and often fail under variations in lighting conditions, hand size, camera angle, and gesture orientation.

Research Question: Can a lightweight DL model reliably recognize gestures from a basic webcam in real-time?

- ***Goal 1:*** Accessibility A hands-free controller anyone can train — alternative to keyboard/mouse.
- ***Goal 2:*** Robustness Prove an MLP on landmarks is far stronger than fragile rule-based methods [1]

Proposed Direction

- **Gesture-controlled Flappy Bird** using an MLP trained on MediaPipe landmarks [12]
- Our MLP is **lightweight and real-time** — unlike heavy CNN-based methods
- **Custom dataset:** Thumbs Up / Thumbs Down /Neutral (≈ 600 samples)
- Addresses the gap **in real-time, low-latency gesture recognition** for practical interactive applications
- **SHAP-based XAI** to validate semantically meaningful learned patterns

Comparative Analysis of Prior Work

Author	Method	Dataset	Acc.	RT?
Zhang (2022)	CNN (5 conv)	ASL Fingerspelling	96.2%	No
Lee (2021)	CNN + LSTM	Chalearn IsoGD	88.7%	No
Chen (2022)	Vision Transformer	Kinect Gesture	95.0%	No
Gupta (2022)	Quantized CNN	Custom IoT	85.0%	Part.
Nguyen (2021)	3D CNN	RWTH-PHOENIX	89.1%	No
Patel (2023)	YOLOv4 + CNN	Custom 3k images	92.0%	No
Köpkü (2019)	ResNeXt-101	EgoGesture	94.0%	Part.
Singh (2023)	CNN classifier	Custom 4-gesture	90.0%	Part.
Ours	MLP + MediaPipe	Custom	99.17%	Yes

RT = Real-Time capable without GPU

Data Collection & Feature Extraction

Tools

- **OpenCV**: live webcam capture
- **MediaPipe**: 21 3D landmarks per frame in real-time

Dataset

- ≈ 600 samples (≈ 200 per class)
- THUMBS UP, THUMBS DOWN, NEUTRAL
- Varied hand position & orientation

Feature Vector

21 landmarks $\times$ 3 coords (x, y, z) :

$$\mathbf{f} = [x_1, y_1, z_1, \dots, x_{21}, y_{21}, z_{21}] \in \mathbb{R}^{63}$$

Preprocessing

- LabelEncoder for class labels
- One-hot via `to_categorical`
- 80/20 train-test split

MLP Architecture

Why MLP over CNN?

- Landmark vectors are tabular — CNN spatial locality not needed
- Far fewer parameters
- Real-time on CPU, no GPU needed [12]

Layer Summary:

- Input: 63-D vector
- Dense(64, ReLU) + Dropout(0.3)
- Dense(128, ReLU) + Dropout(0.3)
- Dense(64, ReLU)
- Output: Dense(3, SoftMax)

Formal Definition

$$\mathbf{h}_1 = \text{ReLU}(\mathbf{W}_1 \mathbf{f} + \mathbf{b}_1), \mathbf{h}_1 \in \mathbb{R}^{64}$$

$$\mathbf{h}'_1 = \text{Dropout}(\mathbf{h}_1, 0.3)$$

$$\mathbf{h}_2 = \text{ReLU}(\mathbf{W}_2 \mathbf{h}'_1 + \mathbf{b}_2), \mathbf{h}_2 \in \mathbb{R}^{128}$$

$$\mathbf{h}'_2 = \text{Dropout}(\mathbf{h}_2, 0.3)$$

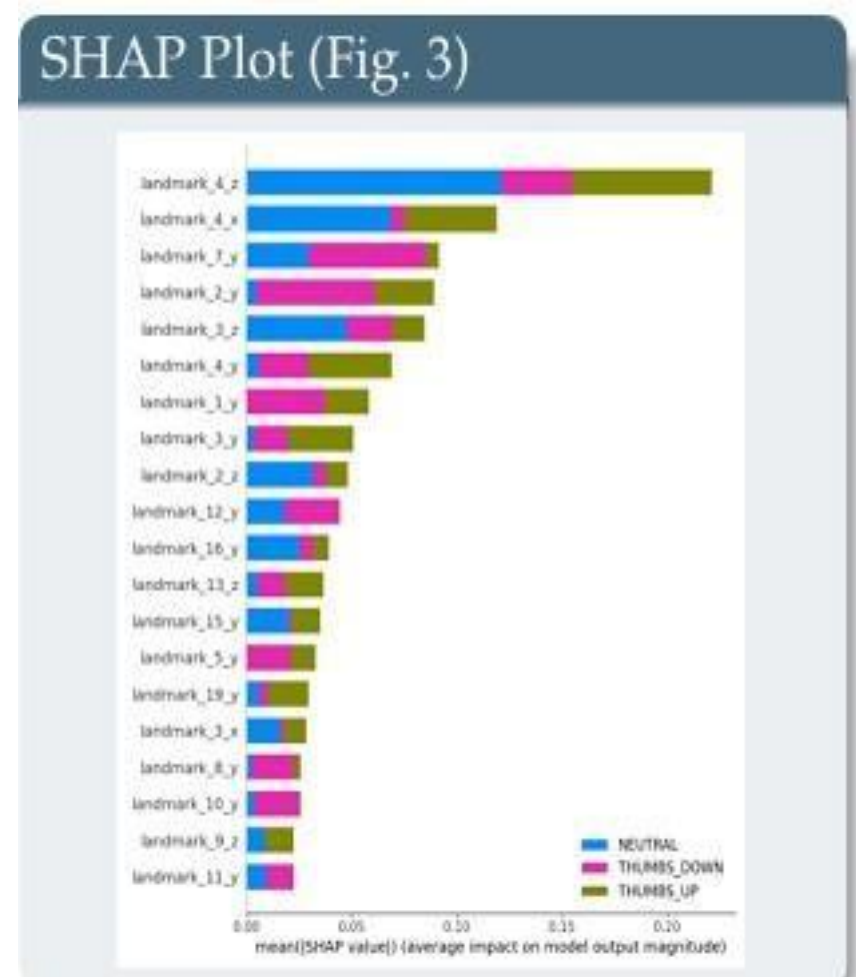
$$\mathbf{h}_3 = \text{ReLU}(\mathbf{W}_3 \mathbf{h}'_2 + \mathbf{b}_3), \mathbf{h}_3 \in \mathbb{R}^{64}$$

$$\hat{\mathbf{y}} = \text{Softmax}(\mathbf{W}_4 \mathbf{h}_3 + \mathbf{b}_4), \hat{\mathbf{y}} \in \mathbb{R}^3$$

$$\mathcal{L} = - \sum_{i=1}^3 y_i \log(\hat{y}_i) \quad (\text{Cross-Entropy Loss})$$

Explainable AI (XAI) — SHAP Analysis

- **SHAP** ranks all 63 features by impact on predictions
- **Top features:**
 - z-coord of **Landmark 4** (thumb tip)
 - Fingertip **Landmarks 8, 12, 16, 20**
- Thumb tip high $\Rightarrow$ Thumbs Up
- Thumb tip low $\Rightarrow$ Thumbs Down
- Model learns **real gesture geometry**, not noise [9]

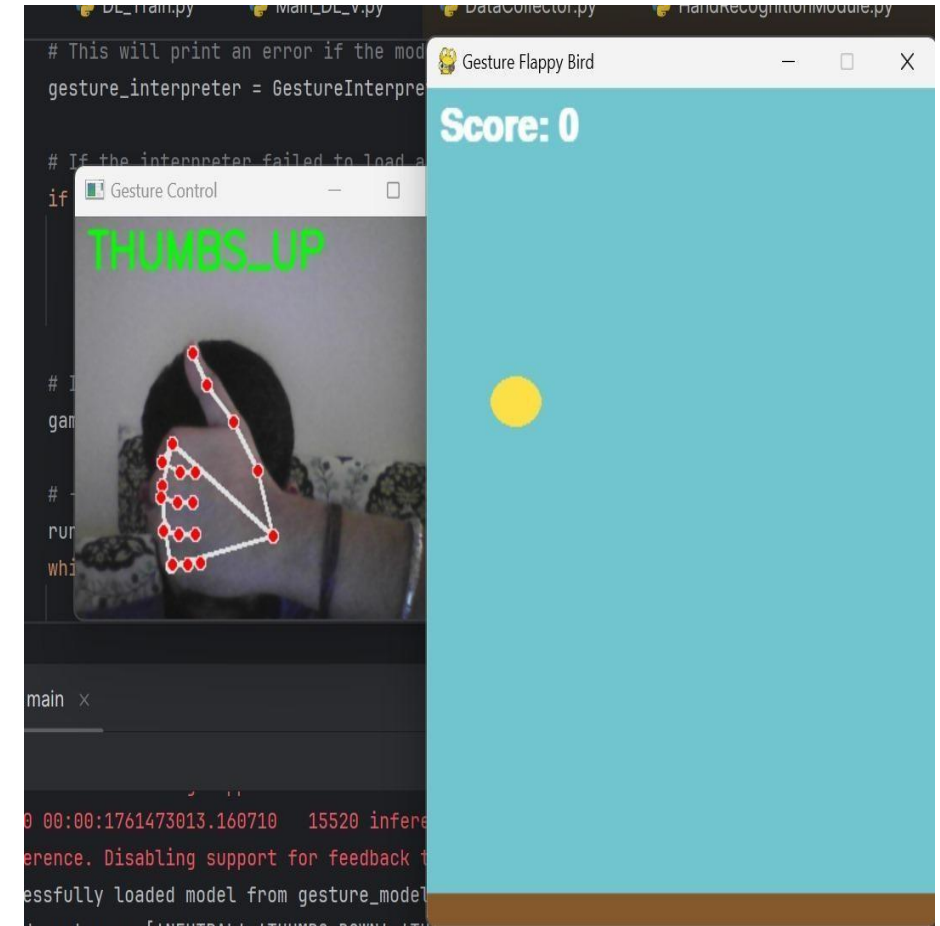


Conclusion and Future Work

What We Demonstrated

- Lightweight MLP on MediaPipe landmarks achieves 99.17% train and 100% test accuracy (120 held-out samples, zero misclassifications)
- Landmark approach outperforms **rule-based methods** and is far more efficient than CNNs
- **SHAP** confirms the model learns real gesture geometry, not spurious patterns
- Full real-time inference on CPU enables practical gesture-controlled gaming.

Future Work: Extend to dynamic gestures (swipe, wave) using lightweight temporal models [10][11][4]



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Code Repository:
<https://github.com/shreyasc2004/Gesture-controlled-flappy-bird-game>

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